

Subject: PRF193 - Programming Fundamentals (with C/C++).

Topic 03: Custom Furniture & Manufacturing Orders

1. Objectives:

Design and implement a C++ application to manage customizable furniture items and the production workflow assigned to craftsmen. The project requires the application of advanced C++ concepts:

- **Object-Oriented Programming (OOP):** Encapsulation, Inheritance, and Polymorphism.
- **Standard Template Library (STL):** Effective use of containers (e.g., `std::vector`, `std::map`) and algorithms.
- **Memory Management:** Proper use of constructors, destructors, and smart pointers.
- **File Processing:** Persistent data storage using text or binary files.
- **Advanced Features:** Exception Handling for robust input validation.

2. Deliverables & Assessment

- **Group Size:** 4 students.
- **Submission Files:**
 - **Source Code:** Fully commented C++ files.
 - **Technical Report:** A PDF explaining the architecture and implementation (class diagram and flow-chart).
 - **Presentation:** Slides covering the core logic and features.
- **Duration:** 09 weeks (from 2nd to 10th).
- **Submission:** via LMS (as directed by the instructor).

3. Functional Requirements

A. Furniture Management

- **Attributes:**
 - Furniture ID (unique)
 - Material Type (Wood, Metal, Plastic)
 - Dimensions

- Base Color
- **Operations:**
 - Add new furniture items
 - Search furniture by ID or material type
 - Update furniture details

B. Order Management

- **Attributes:**
 - Order ID (unique)
 - Assigned Carpenter Name
 - Start Date
 - Estimated Completion Time
- **Operations:**
 - Assign manufacturing tasks to carpenters
 - Calculate pricing based on material area and labor complexity
 - Track orders by carpenter or completion status

C. Data Analysis

- Implement template functions to sort orders by production duration or material cost
- Provide search and filter options for furniture and orders

D. Advanced Logic:

- Use Exception Handling to validate input (e.g., invalid dimensions, missing carpenter ID)
- Utilize STL Maps to group and track furniture items by material type

E. User Interface:

- A professional console-based menu system for easy navigation.
- Design a simple and intuitive text-based menu system.

F. File I/O:

- Load and save data from/to text or binary files.

4. Technical Requirements:

Requirement	Description
Modular Programming	Code must be organized into multiple header (.h) and implementation (.cpp) files.
Data Persistence	Automatically load data from file on startup and save upon exit.
Encapsulation	All class members must be private with appropriate getters, setters, and constructors.
Memory Safety	Use smart pointers (std::unique_ptr or std::shared_ptr) to prevent memory leaks.
Validation	Use try-catch blocks to handle invalid inputs, such as incorrect dimensions, invalid carpenter IDs, or negative material costs.

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